

Name: _____



Making cities greener

Task A: The importance of urban greening

- 1) Imagine you were the mayor of a medium-sized city, and were given the task of adding two different **urban** greening strategies of your choice to the city.
 - a) Which two strategies do you think would be the most cost effective to implement? Explain why.

 - b) Which two strategies do you think would be most effective at improving the well-being of the population of the city? Explain why.

 - c) Which two strategies do you think would be most effective at improving the sustainability of the city? Explain why.

Note: You can use the same strategy for more than one question.

- 2) Several old cities in the UK have tried to implement **urban** greening strategies, but have struggled to do so. Give one possible reason why.

Task B: The maths behind urban greening

- 1) The **urban** city of Oakfield has been given £ 10 000 000 of funding to start an **urban** greening initiative.

transport carbon emissions	461.3 kt
population	102 504
total area of city	98 km ²
area available for renewal	23 km ²
area of bus depot.	7500 m ²
yearly total sunshine hours	1356 hours

planting trees along streets (requires average of 3.5 hours of daily sunshine)	£17 000 per km ²
cycle lanes (suggested only if transport carbon emissions per person are over 0.004 kt)	£4 100 000
urban park renewal (requires 25% of city area to be available)	£35 000 per km ²
electric, zero-emission buses (area of 200 m ² required per bus)	£320 000 per bus

Using the tables of information about Oakfield and the cost of each **urban** greening strategy, recommend a set of effective **urban** greening strategies (and how much of the city it will apply to) that Oakfield can afford to implement.

- 2) Discuss possible limitations of your proposed **urban** greening initiative in question 1.